

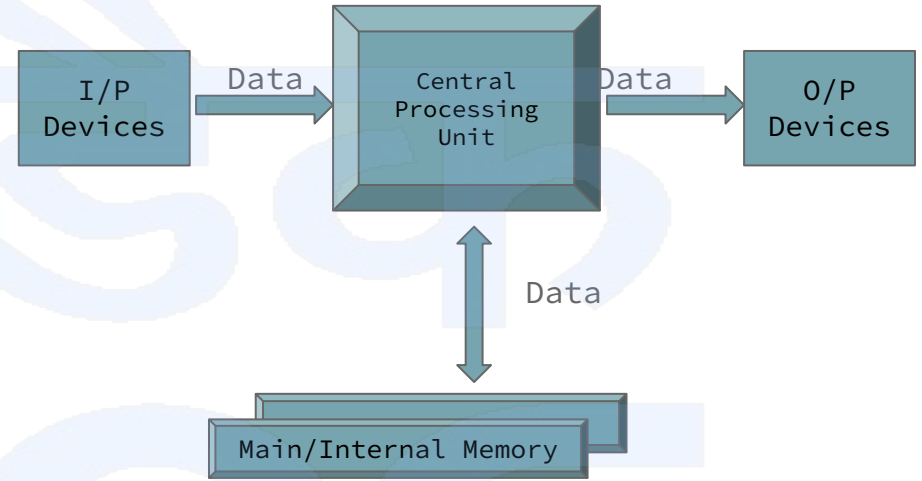


HIGH PERFORMANCE COMPUTING: POWERING THE FUTURE

Presentor

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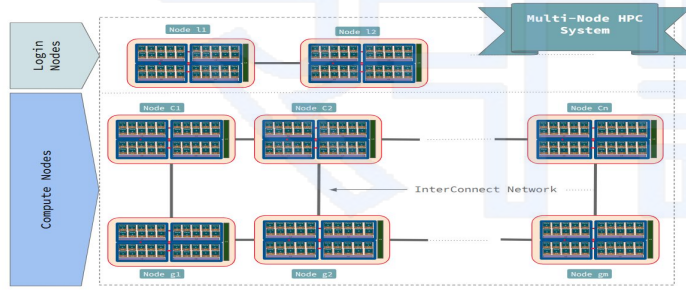


What is computer?

WHAT IS SUPER COMPUTER

OR

HIGH PERFORMANCE COMPUTING?



Ability to process large data and perform complex Calculations at high speeds.

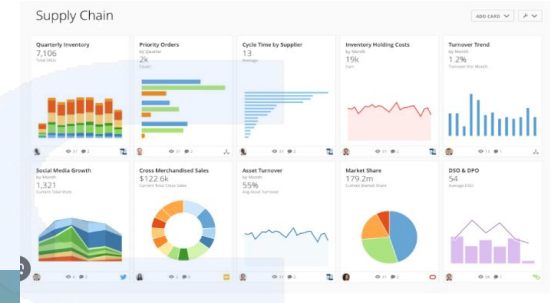
Typically have a large number of nodes having one or more processing chips and connected with very fast network



WHY WE NEED HPC?

Command: BCDEFGIMPRSTUV-

	A	B	C	D
1 Year	1979	1980	1981	
3 Sales	54321	59753	65728	
4 Cost	43457	47802	52583	
5 Profit	10864	11951	13146	
6				
7				
8				
9				
10				
11				
12				

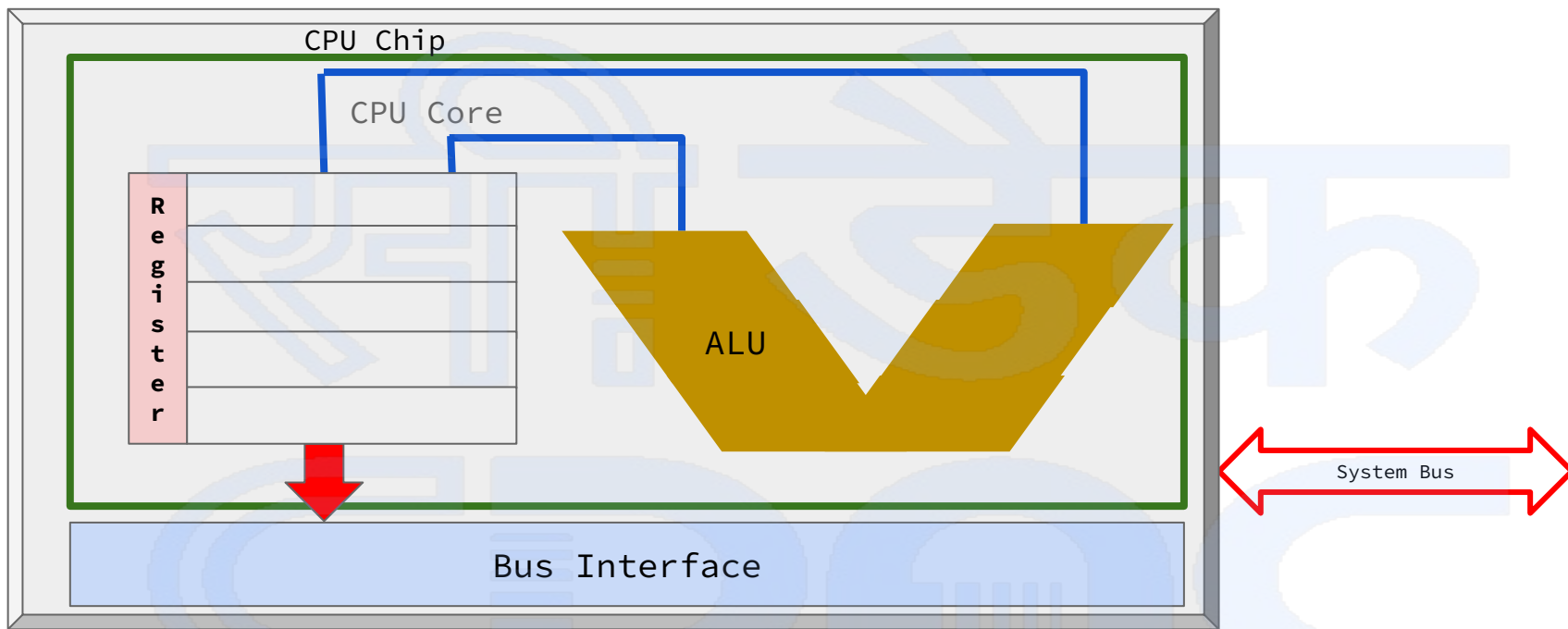


Our programs are becoming more and more complex just like our video games and social media trends

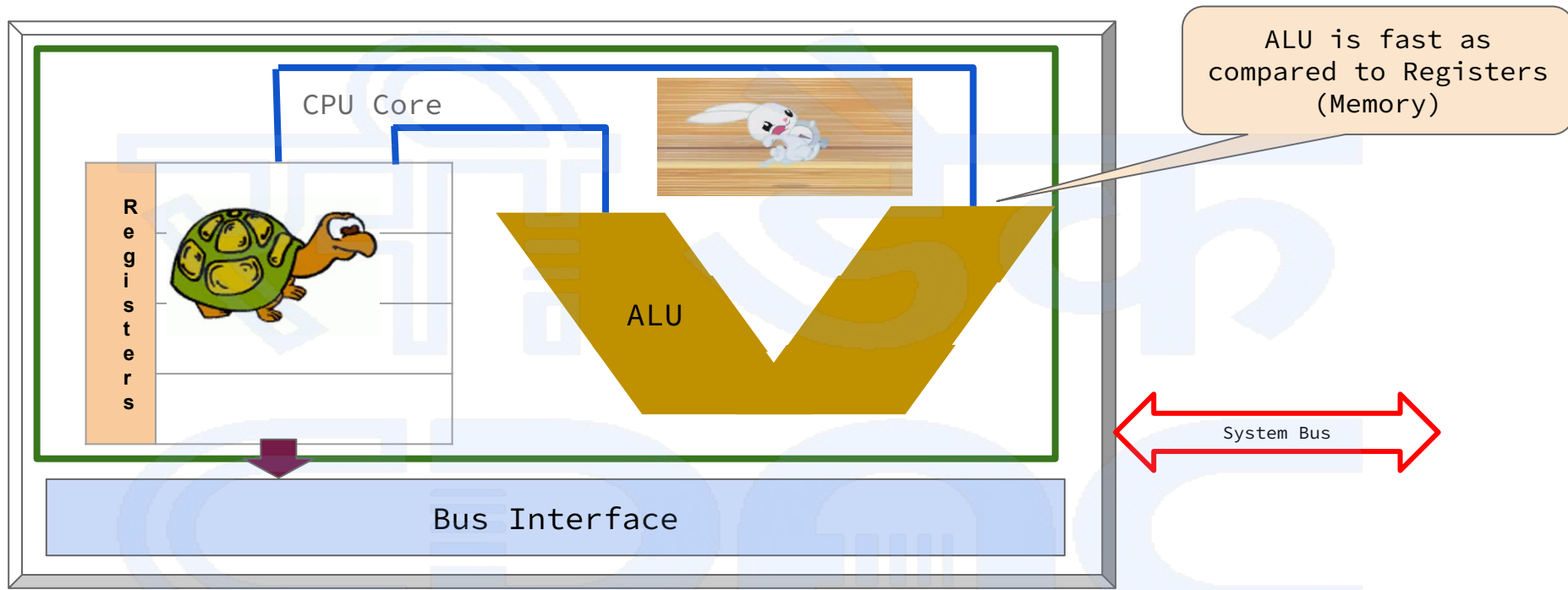


EVOLUTION OF COMPUTING POWER

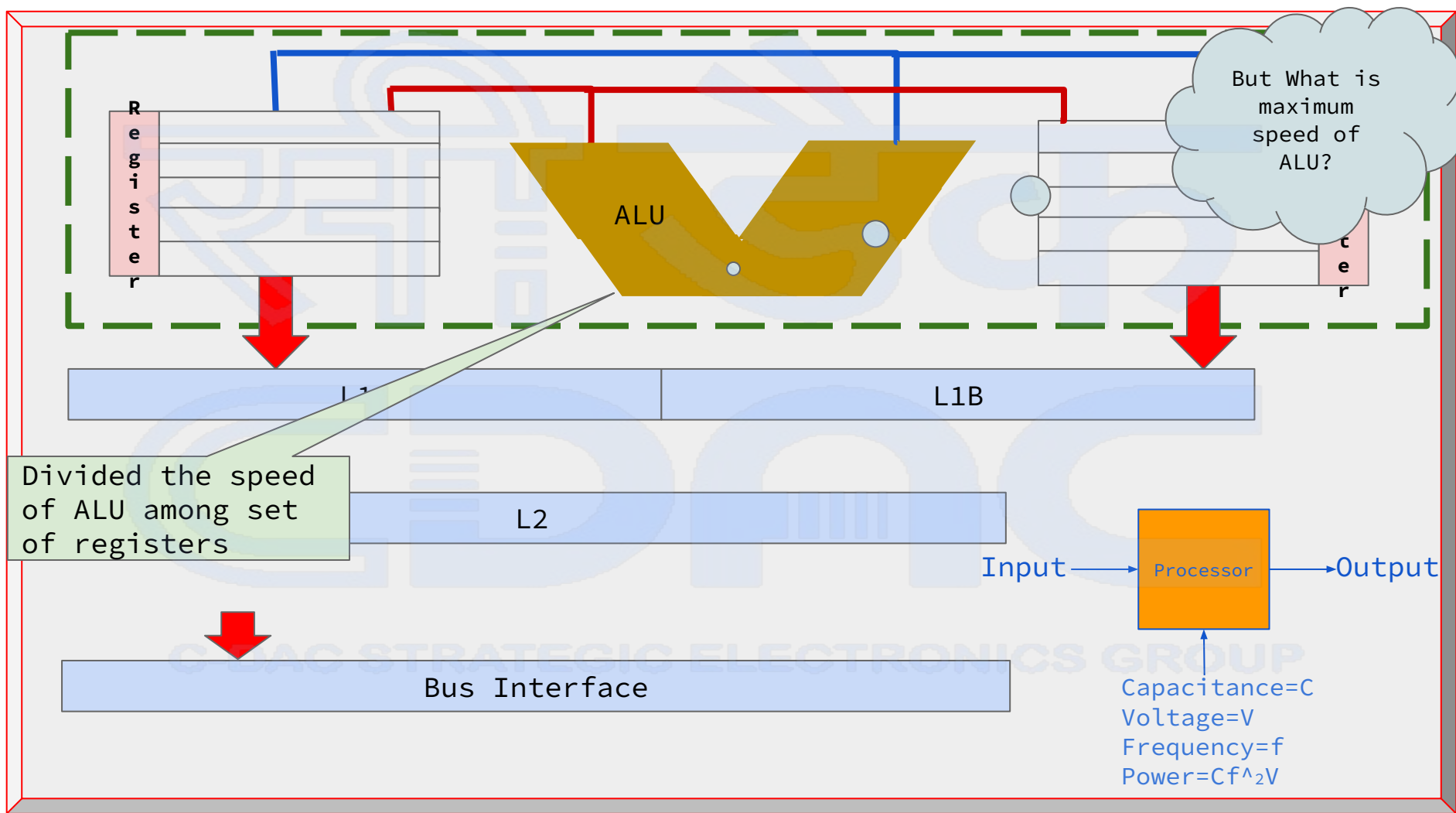
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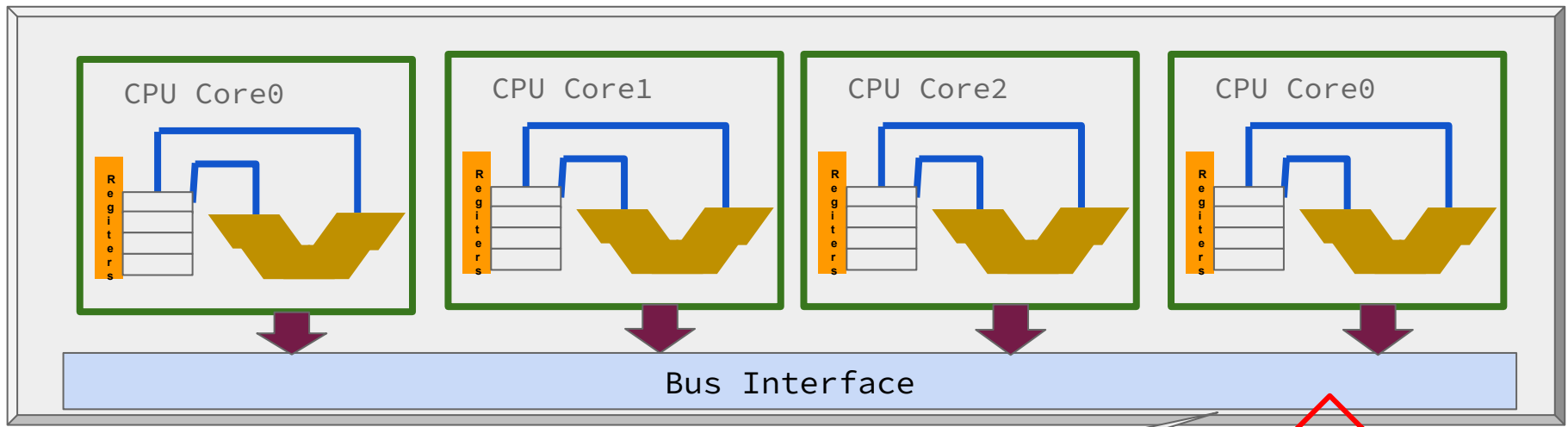


Basic Processor



Different components have different speeds

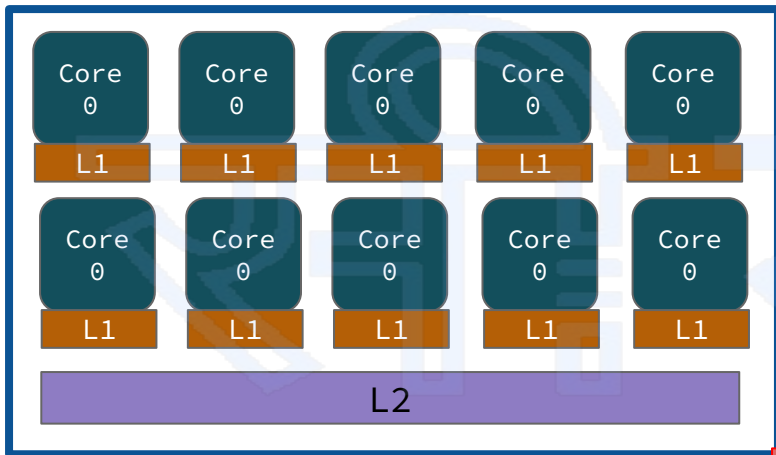




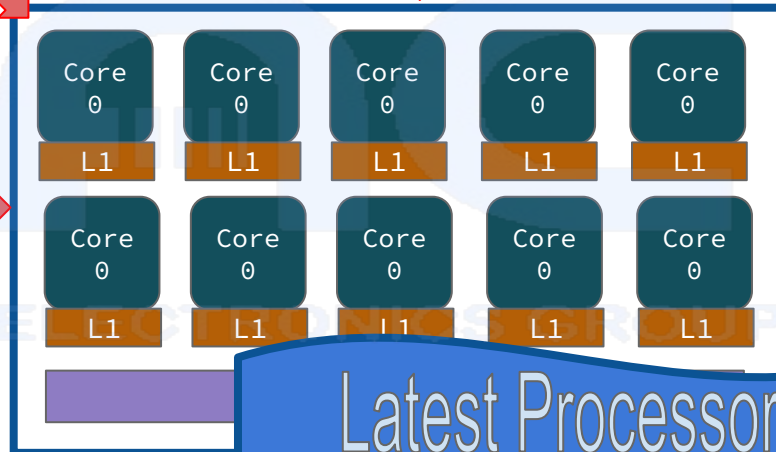
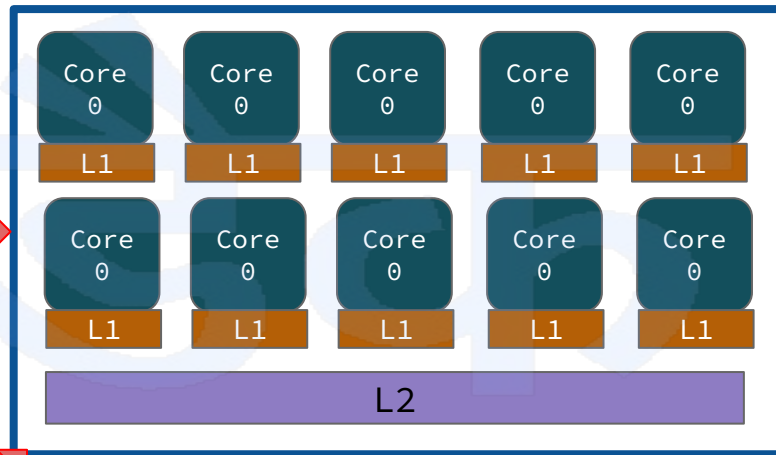
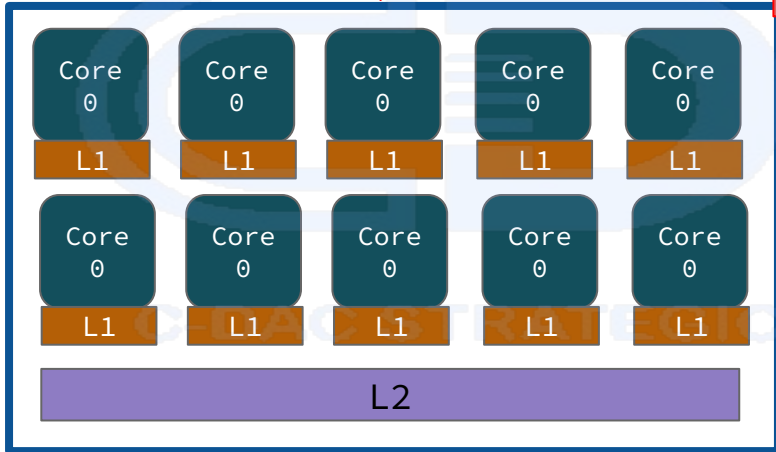
Multiple computing units are integrated on single chip which are called multi core

Multiple Computing Cores

Cores Clusters



Cores Clusters



L3

Latest Processors

WHAT ARE TERAFLUPS?

Measuring Computational Power

Definition: Teraflops (TFLOPS) is a unit of measure for a computer's processing speed, specifically its ability to perform one trillion (10^{12}) floating-point operations per second.

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KEY FACTORS INFLUENCING TFLOPS:

Number of Cores/Processing Units: More units allow for parallel calculations.

Clock Speed (Frequency): How many cycles per second the processor runs (e.g., GHz).

Floating-Point Operations per Cycle: The number of calculations each core/unit can perform in one clock cycle (often leveraging SIMD).

General Formula (Theoretical Peak):

$$\text{TFLOPS} = (\text{Number of Cores} \times \text{Clock Speed (in Hz)} \times \text{Operations per Cycle}) / 10^{12}.$$

CONSIDER A CPU WITH THE FOLLOWING SPECIFICATIONS:

Number of Cores: 16

Clock Speed: 4.5 GHz (which is equal to 4.5×10^9 Hz)

Operations per Cycle (Double Precision): Let's assume each core can perform 8 floating-point operations per cycle in double precision using its vector processing units (like AVX-512, where applicable). This is a high-end assumption for illustrative purposes.

Now, let's plug these values into the TFLOPS formula:

$$\text{TFLOPS} = (\text{Number of Cores} \times \text{Clock Speed (in Hz)} \times \text{Operations per Cycle}) / 10^{12}$$

Step-by-step calculation:

Multiply the number of cores by the clock speed: $16 \times 4.5 \times 10^9 = 72 \times 10^9$

Multiply the result by the operations per cycle: $72 \times 10^9 \times 8 = 576 \times 10^9$

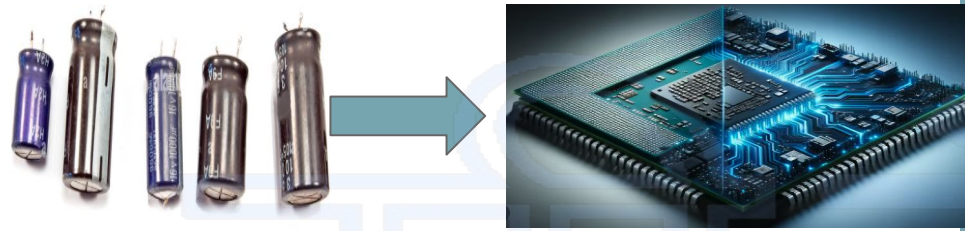
Divide the result by 10^{12} to get Teraflops:

$$\text{TFLOPS} = 576 \times 10^9 / 10^{12} = 0.576$$

IMPORTANT NOTE:

This formula provides the theoretical peak performance. Real-world performance is often lower due to factors like:

- Memory bandwidth limitations
- Software optimization
- Specific types of operations being performed



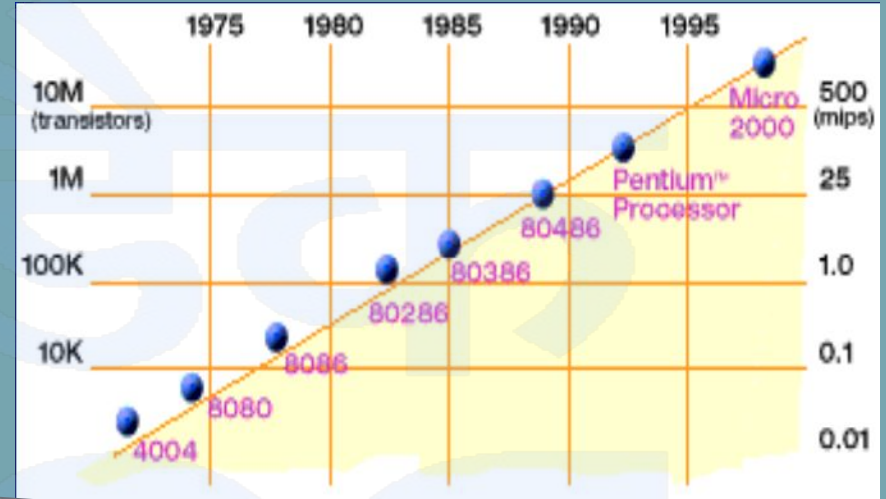
Moores Law

Transistors :

represents the binary digits 0 and 1, foundation of modern computing.

combined to form logic gates, which compare input signals to produce an output.

Logic gates are combined to form complex circuits, which are used to make computer chips

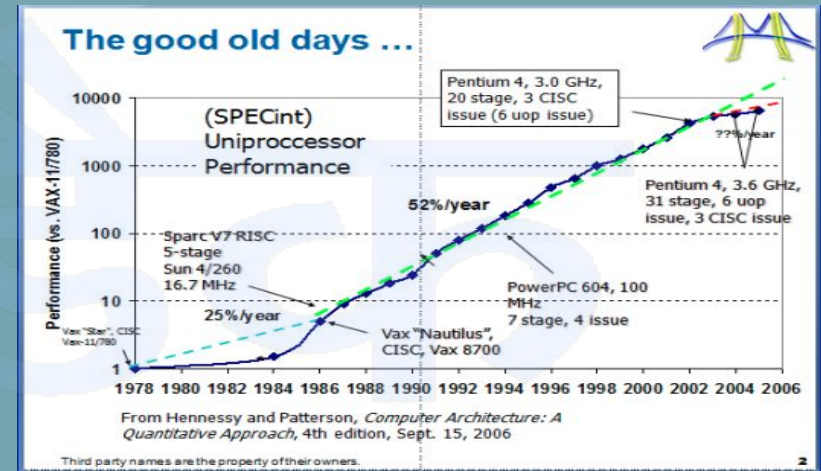


of transistors
microchip doubles every 18
months, leading to exponential growth
in computing power.

Accuracy: The prediction has held true
for decades, with transistors
continuing to shrink and improve.

MOORE'S LAW ... CONSEQUENCES

The Hardware/Software
contract



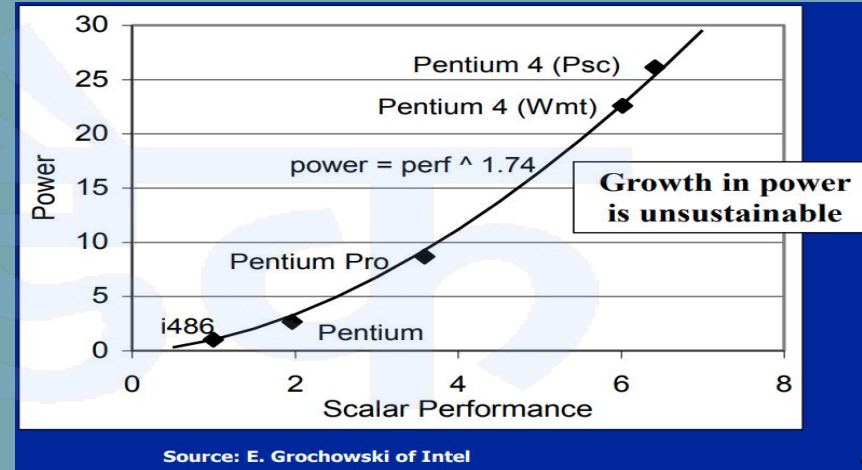
Early software development
relied on hardware improvements
for performance gains.

Many engineers wrote
performance-inefficient code
assuming hardware would always
compensate.

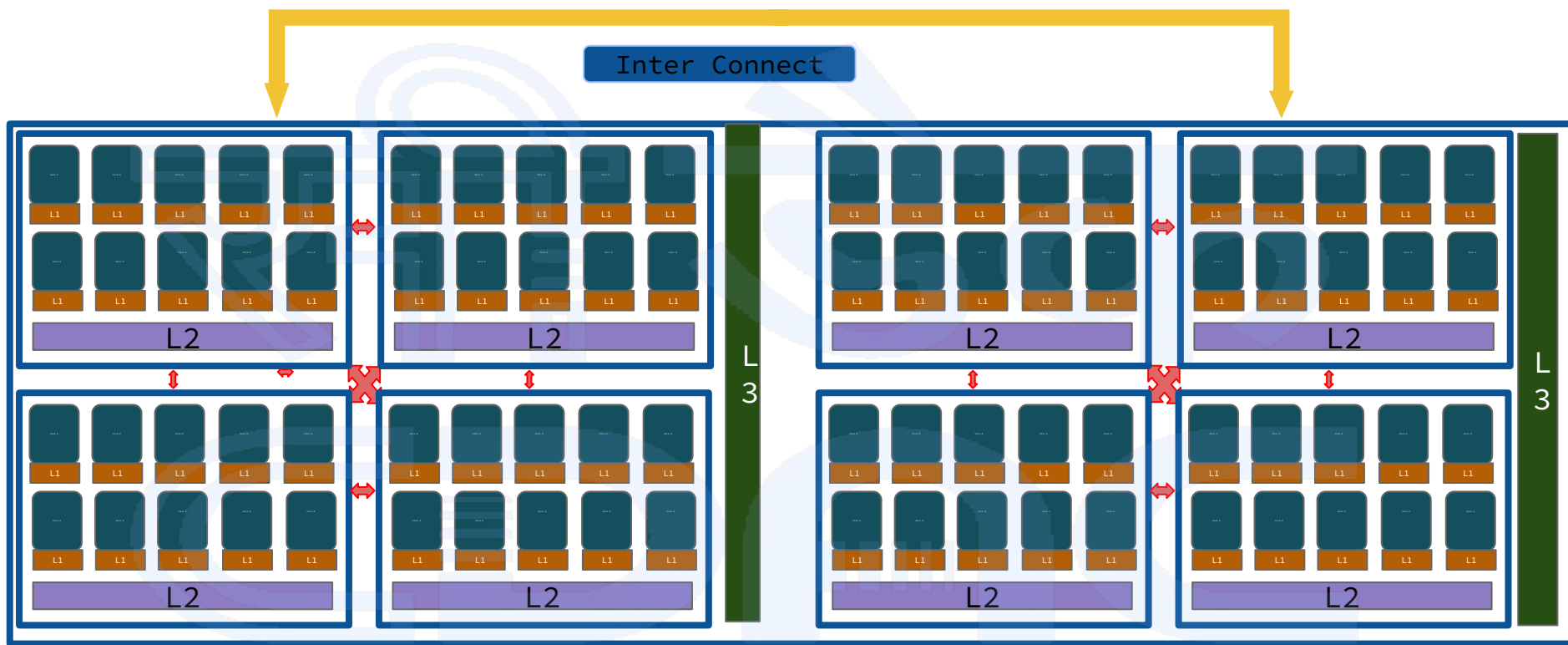
MOORE'S LAW...

POWER WALL

The Hardware/Software
contract



As transistor density increased, power consumption and heat dissipation became unsustainable.



1X

~2X

1X

Joined multiple
nodes with a
fast network

Multi-Node HPC System

Login Nodes

Compute Nodes

Node l1

Node l2

Node C1

Node C2

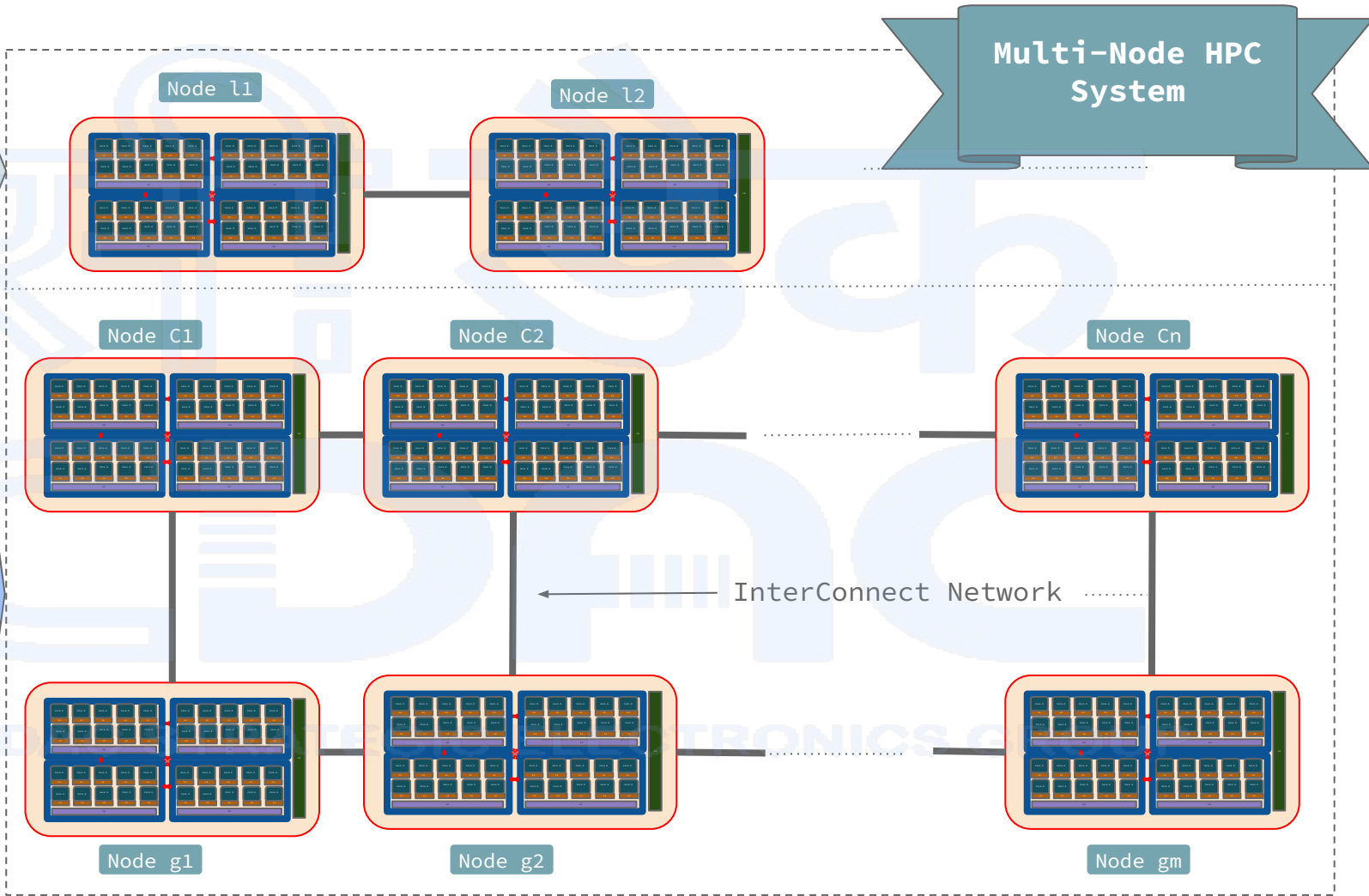
Node Cn

Node g1

Node g2

Node gm

InterConnect Network



MODERN TRENDS

- 1) Shift to Multi-Core and Parallel Computing
 - a) Instead of increasing single-core speed, modern processors feature multiple cores to execute tasks in parallel.
 - b) Technologies like SIMD, SIMT, OpenMP, and MPI enable efficient parallel processing in CPUs and GPUs.
 - c) Example: AMD Ryzen & Intel Core processors now have up to 64 cores for high-performance tasks.
- 2) Rise of Specialized Hardware (Domain-Specific Processors)
 - a) General-purpose CPUs are complemented by GPUs (for AI & graphics), TPUs (for machine learning), and FPGAs (for custom acceleration).
 - b) Example: NVIDIA's Tensor Cores accelerate deep learning, while Google's TPUs power AI workloads in cloud computing.



What is the primary prediction of Moore's Law?

- A) Transistors in a microchip will remain constant over time
- B) The number of transistors in a microchip will double every 18 months
- C) Microchips will become obsolete after 50 years
- D) The performance of software will double every 18

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The number of transistors in a microchip will double every 18 months



Which of the following is NOT a modern trend in computing after Moore's Law?

- A) Multi-core and parallel computing
- B) Use of specialized hardware like GPUs and TPUs
- C) Continued focus on increasing single-core processor speed
- D) Advanced packaging and chiplet architecture

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Continued focus on increasing single-core processor speed

NEXT...

We will learn about :

Parallelism

Job Scheduling and Resource Manager

Methodologies of parallel programming

openMP

openMPI

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